Asynchronous Programming

Credit

- This slide is copied and modified from Fulvio Corno, Luigi De Russis @ Politecnico di Torino

Outline

- Callbacks
- Functional Programming
- Asynchronous Programming
- Database Access with SQLite
- Promises
- async/await

JavaScript: The Definitive Guide, 7th Edition
11.1 Asynchronous Programming with Callbacks

CALLBACKS

Callbacks

https://developer.mozilla.org/en-US/docs/Glossary/Callback_function

- A function passed into another function as an argument
  - Then invoked inside the outer function to complete some kind of routine or action
  - Synchronous
  - Asynchronous
- High-order function

```
1. function logQuote(quote) {
2.     console.log(quote);
3. }
4.
5. function createQuote(quote, callback) {
6.     const myQuote = `Like I always say,
   '${quote}'`;
7.     callback(myQuote);
8. }
9.
10.createQuote("WebApp I rocks!", logQuote);
```

Synchronous Callbacks

- Used in functional programming
  - e.g., providing the sort criteria for array sorting

```
1. let numbers = [4, 2, 5, 1, 3];
2. numbers.sort(function(a, b) {
3.    return a - b;
4. });
5. console.log(numbers);
```

```
1. let numbers = [4, 2, 5, 1, 3];
2. numbers.sort(( a, b ) => a – b);
3. console.log(numbers);
```

Synchronous Callbacks

- Example: filter according to a criteria
  - `filter()` creates a new array with all elements for which the callback returns true

```
1. const market = [
2.    { name: 'GOOG', var: -3.2 },
3.    { name: 'AMZN', var: 2.2 },
4.    { name: 'MSFT', var: -1.8 }
5. ];
6.
7. const bad = market.filter(stock => stock.var < 0);
8. // [ { name: 'GOOG', var: -3.2 }, { name: 'MSFT', var: -1.8 } ]
9.
10.const good = market.filter(stock => stock.var > 0);
11.// [ { name: 'AMZN', var: 2.2 } ]
```

JavaScript: The Definitive Guide, 7th Edition
Chapter 6. Array
Chapter 7.8 Functional Programming

FUNCTIONAL PROGRAMMING

Functional Programming: A Brief Overview

- A programming paradigm where the developer mostly construct and structure code using functions
  - Not JavaScript's main paradigm, but JavaScript is well suited
- More “declarative style” rather than “imperative style” (e.g., for loops)
- Can improve program readability

```
1. new_array =
2. array.filter( filter_function );
```

```
1. new_array = [] ;
2. for (const el of list)
3.    if ( filter_function(el) )
4.       new_array.push(el) ;
```

Notable Features of the Functional Paradigm

- Functions are first-class citizens
  - Functions can be used as if they were `variables` or `constants`, combined with other functions and generate new functions in the process, chained with other functions, etc.
- Higher-order functions
  - A function that operates on functions, taking one or more functions as arguments and typically returning a new function
- Function composition
  - Composing/creating functions to simplify and compress your functions by taking functions as an argument and return an output
- Call chaining
  - Returning a result of the same type of the argument, so that multiple functional operators may be applied consecutively

Functional Programming in JavaScript

- JavaScript supports the features of the paradigm “out of the box”
- Functional programming requires avoiding mutability
  - i.e., do not change objects in place!
  - e.g., if you need to perform a change in an array, return a new array

Iterating over Arrays

- Iterators: `for ... of, for (..;..;..)`
- Iterators: `forEach(f)`
  - Process each element with callback `f`
- Iterators: `every(f), some(f)`
  - Check whether all/some elements in the array satisfy the Boolean callback `f`
- Iterators that return a new array: `map(f), filter(f)`
  - Construct a new array
- `reduce`: callback function on all items to progressively compute a result
  - `reduce(callback(accumulator,currentValue[,index[,array]])[,initialValue])`

Iterables

- `Iterable` objects are a generalization of arrays
  - This concept allows us to make any object useable in a `for..of` loop
- Arrays are `iterable`
  - Many other objects are `iterable` as well. For instance, strings are also `iterable`
- Object represents a collection (`list`, `set`) of something, then `for..of` is a great syntax to loop over it

```
1. let range = {
2.   from: 1,
3.   to: 5
4. };
5.  // We want for(let num of range)
```

- What is a `Iterable`?
  - `Iterable` is an object that defines the Iterator property
  - Iterable is a data structure that makes its elements accessible through iteration
- Iterator is a pointer to the next element in the iteration
  - Iterators often used in combination with the `for...of` loop in JavaScript to iterate over elements

Symbol.iterator

- To make `range` object `iterable`, add a method named `Symbol.iterator`
- 1. When `for..of` starts, it calls that method once or errors if not found
  - The method must return an iterator – an object with the method next
- 3. When `for..of` wants the next value, it calls `next()` on that object.
- Result of `next()` must have the form `{done: Boolean, value: any}`
  - `done=true` means that the loop is finished, otherwise value is the `next` value

```
1. // 1. call to for..of initially calls this
2. range[Symbol.iterator] = function() {
3.  // ...it returns the iterator object:
4.  return {
5.   current: this.from,
6.   last: this.to,
7.   // 3. next() is called on each iteration
8.   next() {
9.    if (this.current <= this.last) {
10.     return {done: false, value: this.current++};
11.   } else {
12.     return { done: true }; // finished
13.   }
14.  }
15. };
16.};
```

Iterables and array-likes

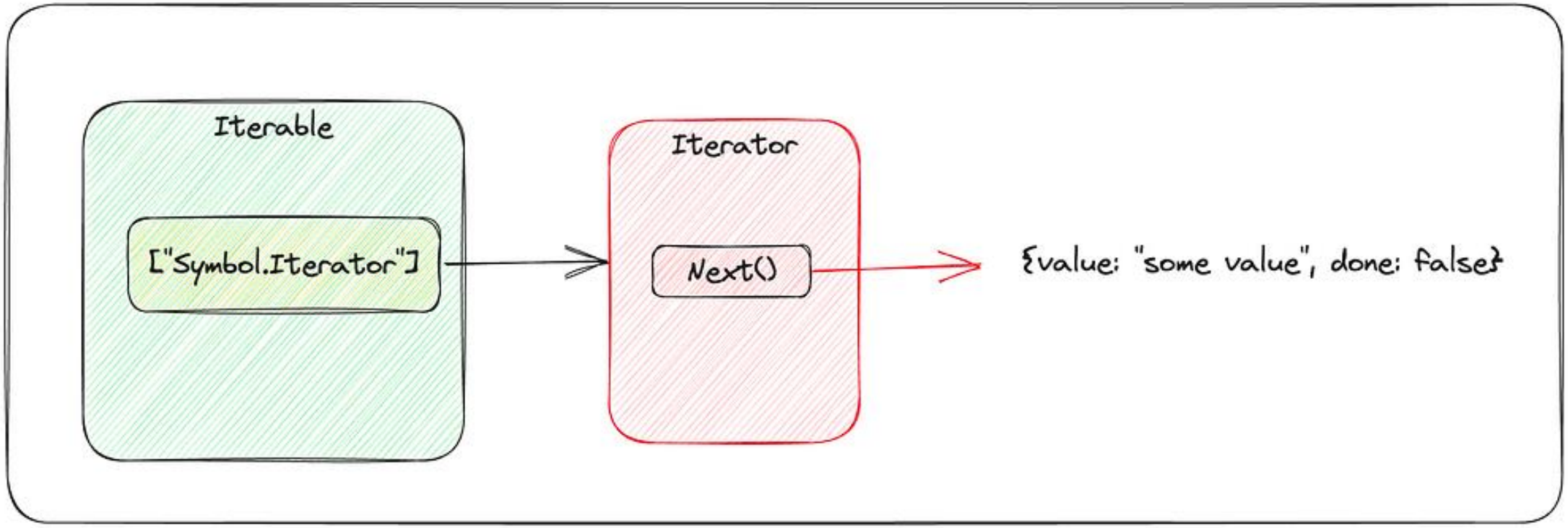
- `Iterables` are objects that implement the `Symbol.iterator` method
- Array-likes are objects that have indexes and length, so they look like arrays
- strings are both `iterable` and array-like
- the `range` in the example above is `iterable`, but not array-like

```
let arrayLike = {
  0: "Hello",
  1: "World",
  length: 2
}; // array-like, not iterable

// Error (no Symbol.iterator)
for (let item of arrayLike) {}
```

- `Array.from`
  - Universal method that takes an `iterable` or array-like value and makes a “real” array

```
let arrayLike = {
  0: "Hello",
  1: "World",
  length: 2
};

let arr = Array.from(arrayLike);
alert(arr.pop()); // World (method works)
```

Iterable and Iterator



.forEach()

https://developer.mozilla.org/en-US/docs/Web/JavaScript/Reference/Global_Objects/Array/forEach

- `forEach()` invokes your (synchronous) callback function once for each element of an `iterable`
  - `forEach(callback(currentElement, index, array), thisValue);`
- The callback may have **3** parameters

```
1. const letters = [..."Hello world"] ;
2. let uppercase = "" ;
3. letters.forEach(letter => {
4.    uppercase += letter.toUpperCase();
5. });
6. console.log(uppercase); // HELLO WORLD
7.
8. let numbers = [1,2,3,4,5];
9. numbers.forEach((num, index, arr) => {
10.   arr[index] = num * 2;
11.   // arr = [2, 4, 6, 8, 10]
12.});
13.console.log(numbers);
```

.forEach()

https://developer.mozilla.org/en-US/docs/Web/JavaScript/Reference/Global_Objects/Array/forEach

- The callback may have **3** parameters
  - `currentValue`: The current element being processed in the array
  - `index` (Optional): The index of `currentValue` in the array
  - `array` (Optional): The array `forEach()` was called upon
- Always returns `undefined` and is not chainable
- No way to stop or break a `forEach()` loop other than by throwing an exception
  - `forEach()` is 95% slower than a traditional `for` loop
- `forEach()` does not mutate the array on which it is called
  - however, its callback may do so

.every()

- `every()` tests whether all elements in the array pass the test implemented by the provided function
- `Callback`: Same 3 arguments as `forEach`
- It returns a **`Boolean`** value (`truthy/falsy`)
- It executes its callback once for each element present in the array until it finds the one where the callback returns a `falsy` value
  - If such an element is found, immediately returns `false`

```
1. let a = [1, 2, 3, 4, 5];
2. a.every(x => x < 10); // => true: all values are < 10
3. a.every(x => x % 2 === 0); // false: not all even values
```

.some()

- `some()` tests whether at least one element in the array passes the test implemented by the provided function
  - It returns a **`Boolean`** value
- It executes its callback once for each element present in the array until it finds the one where the callback returns a `truthy` value
  - if such an element is found, immediately returns true

```
1. let a = [1, 2, 3, 4, 5];
2. a.some(x => x%2 === 0); // => true; a has some even numbers
3. a.some(isNaN);
```

.map()

- **map()** passes each element of the array on which it is invoked to the function you specify
  - Callback should return a value
- **map()** always returns a new array containing the values returned by the callback

```
1. const a = [1, 2, 3];
2. const b = a.map(x => x*x);
3. console.log(b); // [1, 4, 9]
4.
5. const letters = ["Hello world"];
6. const uppercase = letters.map(
7.    letter => letter.toUpperCase()
8. );
9. console.log(uppercase.join(''));
```

.filter()

- **`filter()`** creates a new array with all elements that pass the test implemented by the provided function
  - Callback is a function that returns either true or false
  - If no element passes the test, an empty array is returned

```
1. const a = [5, 4, 3, 2, 1];
2. a.filter(x => x < 3);
3. // generates [2, 1], values less than 3
4. a.filter( (element, index) => index % 2 == 0 );
5. // [5, 3, 1]
```

.reduce()

- `reduce()` **combines** elements of an array, using the specified function, to produce a **single** value
  - A common operation in functional programming and goes by the names "inject" and "fold"

```
1. reduce(
2.    callback(
3.       accumulator,
4.       currentValue[, index[, array]]
5.    )
6.    [, initialValue]
7. }
```

- `reduce()` takes **2** arguments:
  - "reducer function" (callback) that performs reduction/combination operation
    - Combine or reduce 2 values into 1
  - `initialValue` (optional) to pass to the function
    - If not specified, it uses the **first** element of the array as initial value and
    - Iteration starts from the next element

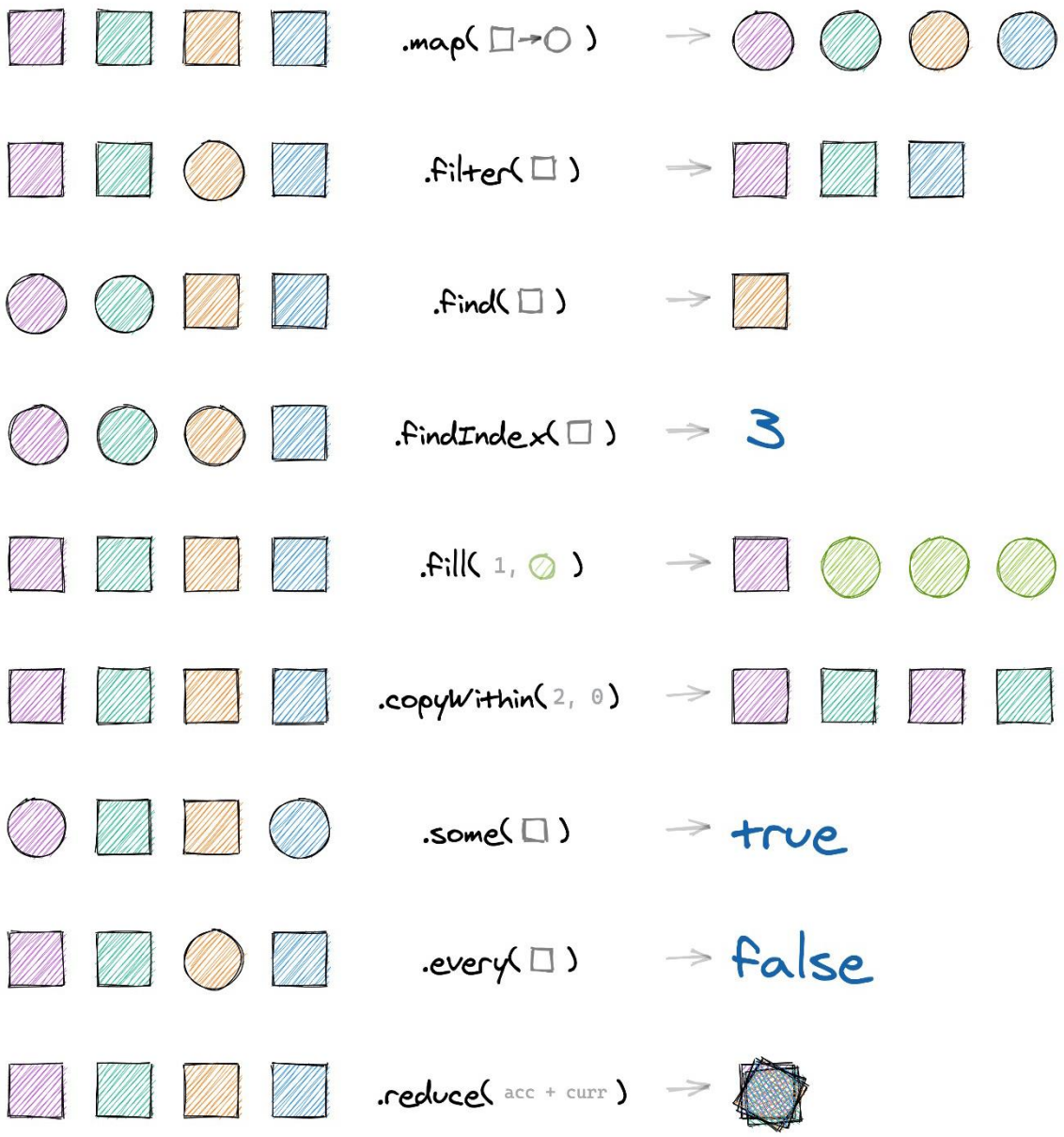
.reduce()

- `Callbacks` used with `reduce()` are different than the ones used with `forEach()` and `map()`
  - First argument is the accumulated result of the reduction so far
  - On the first call to this function, its first argument is the initial value
  - On subsequent calls, it is the value returned by the previous invocation of the reducer function

```
1. const a = [5, 4, 3, 2, 1];
2. a.reduce( (accumulator, currentValue) =>
3.    accumulator + currentValue, 0
4. ); // 15; the sum of the values
5. a.reduce((acc, val) => acc*val,  1);
6.    // 120; the product of the values
7. a.reduce( (acc, val) =>
8.    (acc > val) ? acc : val
9. );
10.   // 5; the largest of the values
```

Array methods cheatsheet

JS tips
@sulco



JavaScript: The Definitive Guide, 7th Edition
Chapter 11. Asynchronous JavaScript

Mozilla Developer Network

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- Web technology for developers » JavaScript » Concurrency model and the event loop
- Web technology for developers » JavaScript » JavaScript Guide » Using Promises

ASYNCHRONOUS PROGRAMMING

Asynchronicity

- JavaScript is **single-threaded** and inherently **synchronous**
  - i.e., code cannot create threads and run in parallel in the JS engine
- Callbacks are the most fundamental way for writing **asynchronous** JS code
- How can they work asynchronously?
  - e.g., how can **`setTimeout()`** or other async callbacks work?
- Thanks to the Execution Environment
  - e.g., browsers and Node.js
- and the Event Loop

```
1. const deleteAfterTimeout = (task) => {
2.    // do something
3. }
4. // runs after 2 seconds
5. setTimeout(deleteAfterTimeout, 2000, task)
```

Non-Blocking Code!

- Asynchronous techniques are useful, particularly for web development
  - e.g., when a web app runs executes an intensive chunk of code without returning control to the browser, the browser can appear to be frozen
  - This is called blocking, and it should be the exception!
    - Browser is blocked from continuing to handle user input and perform other tasks until the web app returns control of the processor
- This may happen outside browsers, as well
  - e.g., reading a long file from the disk/network, accessing a database and returning data, accessing a video stream from a webcam, etc.
- Most of the JS execution environments are, therefore, deeply asynchronous
  - with non-blocking primitives
  - JavaScript programs are event-driven, typically

Asynchronous Callbacks

- The most fundamental way for writing asynchronous JS code
  - Great for “simple” things!
- Handling user actions
  - e.g., button click
- Handling I/O operations
  - e.g., fetch a document
- Handling time intervals
  - e.g., timers
- Interfacing with databases

```
1. const readline = require('readline');
2. const rl = readline.createInterface({
3.    input: process.stdin,
4.    output: process.stdout
5. });
6. rl.question('How old are you? ', (answer) => {
7.    let description = answer;
8.    rl.close();
9. });
10. --------------------
11.let input;
12.rl.on('line', (line) => {
13.   input = line;
14.   rl.close();
15.}
16.console.log(input);
17.undefined
18.> hi
```

Timers

- To delay the execution of a function:
  - `setTimeout()` runs the callback function after a given period of time
  - `setInterval()` runs the callback function periodically

```
1. const onesec = setTimeout(()=> {
2.    console.log('hey') ; // after 1s
3. }, 1000) ;
4. console.log('hi') ;
5.
6. const myFunction = (firstParam, secondParam) => {
7.    // do something
8. }
9. // runs after 2 seconds
10.setTimeout(myFunction, 2000, firstParam, secondParam) ;
```

Timers

- `clearInterval()` for stopping the periodical invocation of `setInterval`

```
1. const id = setInterval(() => {}, 2000) ;
2. // «id» is a handle that refers to the timer
3.
4. clearInterval(id) ;
```

Handling Errors in Callbacks

- No “official” ways, only best practices!
- First parameter of callback function is for storing any error
  - Second one is for the result of the operation
  - this is the strategy adopted by Node.js

```
1. fs.readFile('/file.json', (err, data) => {
2.    if (err !== null) {
3.       console.log(err);
4.       return;
5.    }
6.    //no errors, process data
7.    console.log(data);
8. });
```

PROMISES

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Beware: *Callback Hell*!

- Must keep passing new functions to perform multiple asynchronous actions in a row using callbacks
  - Every callback adds a level of nesting
  - When you have lots of callbacks, code starts to be complicated very quickly
- I am requesting some data from the server by calling `getData()`
  - Inside I am requesting for some more data by calling `getMoreData()` passing previously received data as an argument

```
1. getData(function(x){
2.     console.log(x);
3.     getMoreData(x, function(y){
4.         console.log(y);
5.         getSomeMoreData(y, function(z){
6.             console.log(z);
7.         });
8.     });
9. });
```

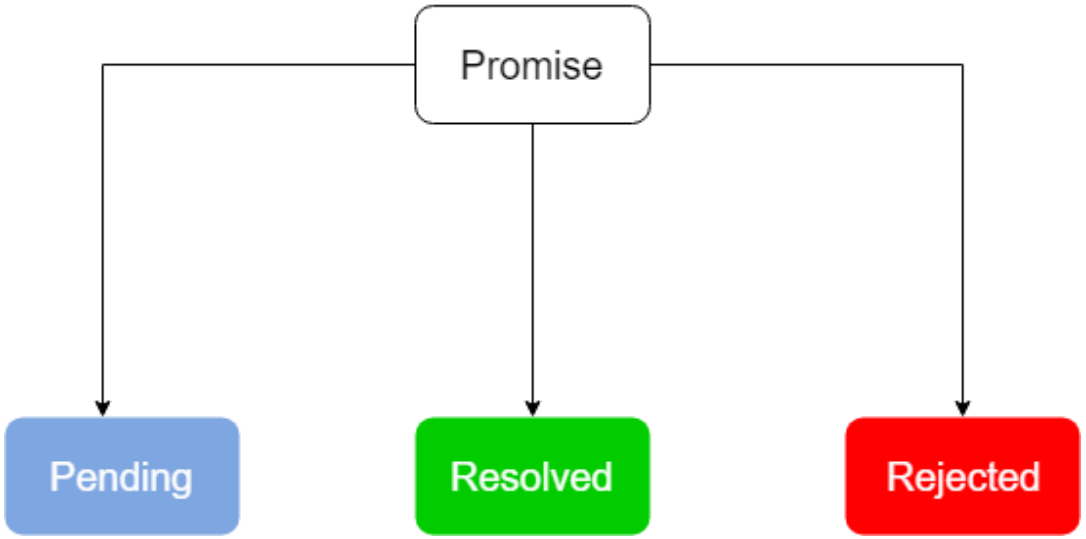
```
1. getData()      // Promise
2.   .then((x) => {
3.     console.log(x);
4.     return getMoreData(x);
5.   })
6.   .then((y) => {
7.     console.log(y);
8.     return getSomeMoreData(y);
9.   })
10.  .then((z) => {
11.    console.log(z);
12.  });
```

Promises

- A core language feature to “simplify” asynchronous programming
  - Promise is an object that holds the future value of an async operation
    - If we are requesting some data from a server, Promise promises us to get that data which we can use in future
- It is an object representing the eventual completion (or failure) of an asynchronous operation
  - i.e., an asynchronous function returns a promise to supply the value at some point in the future, instead of returning immediately a final value
- Promises standardize a way to handle errors and provide a way for errors to propagate correctly through a chain of promises

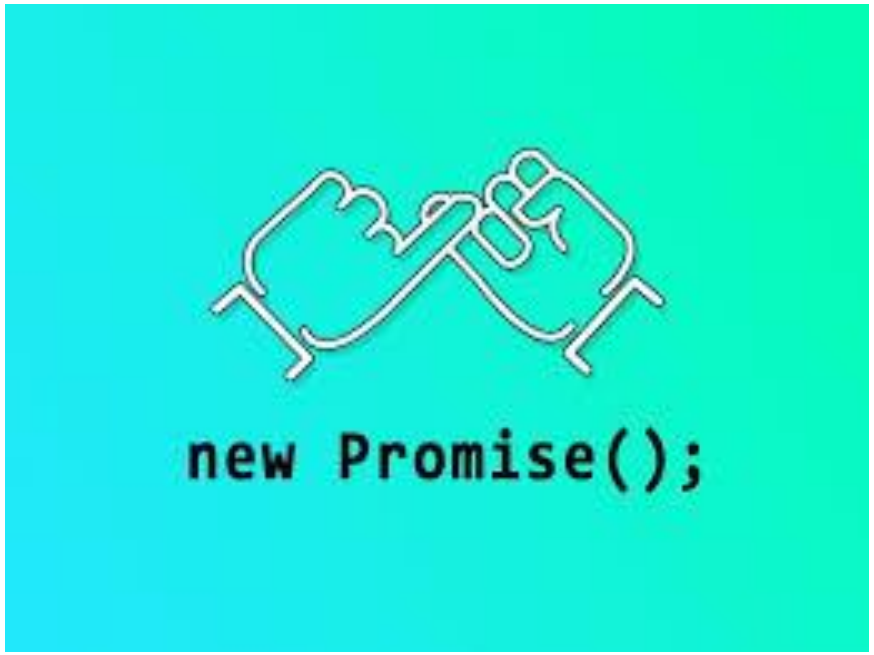
States of a Promise

- Unresolved or Pending
  - Result is not ready
  - Waiting for something to finish (for example, an async operation)
- Resolved or Fulfilled
  - A Promise is resolved if the result is available
  - Something finished and all went well.
- Rejected
  - A Promise is rejected if an error happened



Promises

- Promises can be created or consumed
  - many Web APIs expose Promises to be consumed!
- When consumed:
  - Promise starts in a pending state
    - Caller function continues the execution, while it waits for the Promise to do its own processing, and give the caller function some “responses”
  - Caller function waits for it to either return the promise in a fulfilled state
    or in a rejected state



Creating a Promise

- Created using the **new** keyword with the Promise constructor
  - It takes a single argument, a callback, also known as **executor** function which takes two callbacks, resolve and reject
  - The executor function is immediately executed when a promise is created
- Takes two functions as parameters:
  - **resolve**, called when the asynchronous task completes successfully and returns the results of the task as a value
  - **reject**, called when the task fails and returns the reason for failure (an error object, typically)

```
1. const promise =
2.    new Promise((resolve, reject) => {
3.    const randomNumber = Math.random();
4.    setTimeout(() => {
5.       if(randomNumber < .6) {
6.          resolve('All things went well!');
7.       } else {
8.          reject('Something went wrong');
9.       }
10.   }, 2000);
11.});
```

Promise.resolve( )

- `Promise.resolve()` static method "resolves" a given value to Promise
  - If the value is a promise, that promise is returned
  - if the value is thenable it will call `then()` method with two callbacks
    - Otherwise the returned promise will be fulfilled with the value

```
1. const p = Promise.resolve([1, 2, 3]);
2. p.then((v) => {
3.    console.log(v[0]); // 1
4. });
```

Example

- Promise is resolved or rejected after 2 seconds
  - Resolved if the `randomNumber` is less than .6 and rejected otherwise
- When Promise is created, it will be in the pending state, its value will be undefined
- After the 2 seconds, Promise is either resolved or rejected state
  - Its value will be the value passed to the resolve or reject function
- Executor calls only one resolve or one reject
  - All further calls of resolve and reject are ignored

```
1. const promise =
2.    new Promise((resolve, reject) => {
3.    const randomNumber = Math.random();
4.    setTimeout(() => {
5.       if(randomNumber < .6) {
6.          resolve('All things went well!');
7.       } else {
8.          reject('Something went wrong');
9.       }
10.   }, 2000);
11.});
```

Consuming a Promise

- Promise serves as a link between the executor and the consuming functions
  - Consuming functions can be registered using **`then()`** and **`catch()`** methods

`.then(): promise.then(successCB, failCB)`

  - Promise is resolved, the **`successCB`** is called with the value passed to **`resolve()`**
  - Promise is rejected, the **`failCB`** is called with the value passed to **`reject()`**

```
1. const promise =
2.   new Promise((resolve, reject) => {
3.   const randomNumber = Math.random();
4.
5.   if(randomNumber < .7) {
6.     resolve('All things went well!');
7.   } else {
8.     reject(new Error('Something went wrong'));
9.   }
10.});
11.
12.promise.then((data) => {
13.  console.log(data);
14.   // prints 'All things went well!'
15.  }, // sucessCB
16.  (error) => {
17.  console.log(error); // prints Error object
18.  } // failCB
19.);
```

Consuming a Promise

`.catch(): promise.catch(failCB)`

- Use **`catch()`** for handling errors
  - More readable than handling errors inside the **`failCB`** of the **`then()`** callback

```
1. const promise = new Promise((resolve, reject) => {
2.   reject(new Error('Something went wrong'));
3. });
4. promise
5.   .then((data) => {
6.     console.log(data);
7.    })
8.   .catch((error) => {
9.     console.log(error); // prints Error object
10.  });
```

- You can omit **`catch()`**, if you are interested in the result, only
- If we're interested only in errors, we can use **`null`** as the first argument
  - `.then(null, errorFunction)`
- or can use
  - `.catch(errorFunction)`

Consuming a Promise

- **`p.then(onFulfilled[, onRejected]);`**
  - Callbacks are executed asynchronously (inserted in the event loop) when the promise is either fulfilled (success) or rejected (optional)
- **`p.catch(onRejected);`**
  - Callback is executed asynchronously (inserted in event loop) when promise is rejected
  - Short for **`p.then(null, failureCallback)`**
- **`p.finally(onFinally);`**
  - Callback is executed in any case, when the promise is either fulfilled or rejected
  - Useful to avoid code duplication in then and catch handlers
- All these methods return Promises, too! ⇒ They can be chained

Promise: Create & Consume

```
const prom = new Promise(
   (resolve, reject) => {
      ...
      resolve(x);
      ...
      reject(y);
      ...
  }
)
```

```
prom
   .then((x) => {
      ...use x...
   } )
   .catch( (y) => {
      ...use y...
   } ) ;
```

Promise Chaining

- Chaining resolves promises in a sequence
- Resolving `promise1` call `then()` which returns `promise2`
- Next `then()` is called which returns `promise3`
- `Promise3` is rejected
  - `catch()` is called which handles the `promise3` rejection
  - Generally only one `catch()` is enough for handling rejection

```
1. const promise1 = new Promise((resolve, reject) => {
2.   resolve('Promise1 resolved');
3. });
4. const promise2 = new Promise((resolve, reject) => {
5.   resolve('Promise2 resolved');
6. });
7. const promise3 = new Promise((resolve, reject) => {
8.   reject('Promise3 rejected');
9. });
10.promise1
11.   .then((data) => {
12.     console.log(data);  // Promise1 resolved
13.     return promise2;
14.   })
15.   .then((data) => {
16.     console.log(data);  // Promise2 resolved
17.     return promise3;
18.   })
19.   .then((data) => {
20.     console.log(data);
21.   })
22.   .catch((error) => {
23.     console.log(error);  // Promise3 rejected
24.   });
```

Chaining Promises

- They provide a natural way to express a sequence of asynchronous operations as a linear chain of **`then()`** invocations
  - **Without having to nest** each operation within the callback of the previous one
    - The "callback hell" seen before
  - If error handling code is same for all steps, you can attach **`catch()`** to **end of chain**
- Important: always return results, otherwise callbacks won't get the result of a previous promise

Promise.all()

- It takes an array of promises as input and returns a new promise
  - Fulfills when all of the promises inside the input array have fulfilled or
  - Rejects as soon as one of the promises in the array rejects
- Data inside `then()` is an array which contains promise values in the same order as defined in promise array passed to `Promise.all()`
  - `Promise1 resolved Promise2 resolved`

```
1. const promise1 = new Promise((resolve, reject) => {
2.     setTimeout(() => {
3.         resolve('Promise1 resolved');
4.     }, 2000);
5. });
6. const promise2 = new Promise((resolve, reject) => {
7.     setTimeout(() => {
8.         resolve('Promise2 resolved');
9.     }, 1500);
10.});
11.Promise.all([promise1, promise2])
12.    .then((data) => console.log(data[0], data[1]))
13.    .catch((error) => console.log(error));
```

Promise.all()

- Promise is rejected with the reason of rejection from the first rejected promise in the input array
- As soon as the second promise is rejected after 1.5 seconds, the returned promise from `Promise.all()` is rejected without waiting for the first promise
- So `Promise.all()` waits for all promises to succeed and fails if any of the promises in the array fails

```
1. const promise1 = new Promise((resolve, reject) => {
2.     setTimeout(() => {
3.         resolve('Promise1 resolved');
4.     }, 2000);
5. });
6. const promise2 = new Promise((resolve, reject) => {
7.     setTimeout(() => {
8.         reject('Promise2 rejected');
9.     }, 1500);
10.});
11.Promise.all([promise1, promise2])
12.    .then((data) => console.log(data[0], data[1]))
13.    .catch((error) => console.log(error));
```

Promise.race()

- It takes an array of promises as input and return new promise that
  - Fulfills as soon as one of the promises in the input array fulfills or
  - Rejects as soon as one of the promises in the input array rejects
- As soon as the first promise is resolved after 1 second
  - Returned promise from `Promise.race()` is resolved without waiting for the second promise to be resolved or rejected

```
1. const promise1 = new Promise((resolve, reject) => {
2.     setTimeout(() => {
3.     resolve('Promise1 resolved');
4.     }, 1000);
5. });
6. const promise2 = new Promise((resolve, reject) => {
7.     setTimeout(() => {
8.         reject('Promise2 rejected');
9.     }, 1500);
10.});
11.Promise.race([promise1, promise2])
12.  .then((data) => console.log(data))  // resolved
13.  .catch((error) => console.log(error));
```

Example: Chaining

- Useful, for instance, with I/O API such as **`fetch()`**, which returns a Promise

```
1. const status = (response) => {
2.    if (response.status >= 200 && response.status < 300) {
3.       return Promise.resolve(response)
4.       // static method to return a fulfilled Promise
5.    }
6.    return Promise.reject(new Error(response.statusText))
7. }
8. const json = (response) => response.json()

9. fetch('/todos.json’)
10.   .then(status)
11.   .then(json)
12.   .then((data) => { console.log('Request succeeded with JSON response', data) })
13.   .catch((error) => { console.log('Request failed', error) })
```

ASYNC/AWAIT

JavaScript: The Definitive Guide, 7th Edition
Chapter 11. Asynchronous JavaScript

Mozilla Developer Network

- Learn web development JavaScript » Dynamic client-side scripting » Asynchronous JavaScript
- Web technology for developers » JavaScript » Concurrency model and the event loop
- Web technology for developers » JavaScript » JavaScript Guide » Using Promises

How It Works

- `async/await` syntax is a special syntax created to help you work with `promise`
  - Need to chain calls to function or variable that returns a `Promise` using `then`/`catch` methods
- When you have many promises, you will also need lots of then method chains

- `fetch()` function returns a Promise, which we handle using `then()` method
  - First `then()`, we accept the response from the request and convert it into an object
  - Second `then()`, we receive the returned json then log that data to the console.
  - `Catch()` method handles any error that might happen

```
1. fetch('https://jsonplaceholder.typicode.com/todos/1')
2.   .then(response => response.json())
3.   .then(json => console.log(json))
4.   .catch(error => console.log(error));
```

Fetch API

- Simple interface for fetching resources
- It allows us to make network request and handle responses easier than our old `XMLHttpRequest(XHR)`
  - Fetch API uses Promises, which provides a way to avoid callbacks hell and boilerplate heavy code that XHR provides

- Promise, returned by fetch, resolves with an object of the built-in Response class

```
1. let promise = fetch(url, [options])
```

Await

```
1. fetch('https://jsonplaceholder.typicode.com/todos/1')
2.   .then(response => response.json())
3.   .then(json => console.log(json))
4.   .catch(error => console.log(error));
```

- `await` makes JavaScript **wait** until the Promise object is resolved or rejected
  - Instead of having to use the callback pattern inside `then()`, you can assign the fulfilled promise into a variable

```
1. const response = await
   fetch('https://jsonplaceholder.typicode.com/todos/1');
2. const json = await response.json();
3. console.log(json)
```

- `await` keyword is placed before call to a function or variable that returns promise
  - It makes JavaScript wait for promise object to settle before running code in next line

- SyntaxError
  - `await is only valid in async functions and the top level bodies of modules`

Simplifying Writing With **`async/await`**

- ECMAScript 2017 (ES8) introduces two new keywords, **`async`** and **`await`**
  - Write promise-based asynchronous code that looks like synchronous code
- Prepend **`async`** keyword to any function means that it will return **`Promise`**
- Prepend **`await`** when calling an **`async`** function (or a function returning a **`Promise`**) makes the calling code stop until promise is resolved or rejected

```
1. const sampleFunction = async () => {
2.    return 'test'
3. }
4. sampleFunction().then(console.log)
5. // This will log 'test'
```

Async

- You add `async` keyword before function name to create asynchronous function
- We created an asynchronous function called `runProcess()` and put the code that uses the `await` keyword inside it
  - We can then run the asynchronous function by calling it

```
1. async function runProcess() {
2.    const response = await
3.    fetch('https://jsonplaceholder.typicode.com/todos/1');
4.    const json = await response.json();
5.    console.log(json)
6. }
7.
8. runProcess();
```

How to Handle Errors in `Async/Await`

- You can use the `try/catch` block to catch any rejection from your promise
- `try/catch` block, put inside the `runProcess()` function, will handle the rejection that comes from the promise

```
1. async function runProcess() {
2.    try {
3.       const response = await
4.       fetch('https://jsonplaceholder.typicode.com/todos/1');
5.       const json = await response.json();
6.       console.log(json);
7.    } catch (error) {
8.       console.log(error);
9.    }
10.}
11.runProcess();
```

PROMISE

```
fetch('https://jsonplaceholder.typicode.com/todos/1')
  .then(response => response.json())
  .then(json => console.log(json))
  .catch(error => console.log(error));
```

ASYNC/AWAIT

```
async function runProcess() {
  try {
    const response = await fetch('https://jsonplaceholder.typicode.com/todos/1');
    const json = await response.json();
    console.log(json);
  } catch (error) {
    console.log(error);
  }
}

runProcess();
```

Examples: Before and After

```
1. const makeRequest = () => {
2.    return getAPIData()
3.     .then(data => {
4.       console.log(data);
5.       return "done";
6.       }
7.    );
8. }
9.
10.let res = makeRequest();
```

- Arrow function

```
1. const makeRequest = async () => {
2.    console.log(await getAPIData());
3.    return "done";
4. };
5.
6. let res = makeRequest();
```

Examples: Before and After

```
1. function getData() {
2.    return getIssue()
3.       .then(issue => getOwner(issue.ownerId))
4.       .then(owner => sendEmail(owner.email, 'Some text'));
5. }
6. // assuming that all the 3 functions above return a Promise
```

```
1. async function getData = {
2.    const issue = await getIssue();
3.    const owner = await getOwner(issue.ownerId);
4.    await sendEmail(owner.email, 'Some text');
5. }
```

Chaining with async/await

- Simpler to read, easier to debug
  - Debugger would not stop on asynchronous code

```
1. const getFirstUserData = async () => {
2.    const response = await fetch('/users.json'); // get users list
3.    const users = await response.json(); // parse JSON
4.    const user = users[0]; // pick first user
5.    const userResponse = await fetch(`/users/${user.name}`); // get user data
6.    const userData = await user.json(); // parse JSON
7.    return userData;
8. }
9. getFirstUserData();
```

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